

Solution Requirements

Date	30 SEPTEMBER 2026
Team ID	SWTID-2026-4644
Project Name	FIT BUDDY
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Selects a beginner routine, follows the guided exercises, and logs the completed session.</i>	Medium
2	Authorization levels	<i>Energized, physically challenged, and accomplished.</i>	High
3	External interfaces	<i>Google gemini api key can be used in this fit buddy project</i>	High

4	Transactions processing	<i>Securely process monthly and annual subscription payments through third-party gateways. Automate database updates to instantly elevate a user's role to "Premium" upon successful transaction confirmation, and handle automatic cycle renewals or cancellations without manual administrative intervention.</i>	High
5	Reporting	<i>User-Facing: Real-time dashboards displaying daily streak counts, weekly workout completion rates, and historical performance charts. Ability to export 12-week progress summaries as a PDF.</i> <i>Admin-Facing: System must generate automated weekly product reports tracking Daily Active Users (DAU), subscription churn rates, and feature-usage heatmaps to inform product marketing decisions</i>	High

6	Business rules, etc.	<i>A user's 7-day free trial automatically downgrades to a "Free" tier profile unless a valid payment method is successfully charged on day 8.</i> <i>• Workout streak logic: A daily streak is only incremented if the user completes at least 80% of the active workout timer; users cannot manually override completed workout durations to manipulate leaderboards.</i> <i>• Account deletion requests must permanently purge all associated biometric and progress data within 30 days.</i>	Medium
7	Compliance to laws or regulations	<i>System architecture and data handling must strictly adhere to GDPR and CCPA regulations, requiring explicit user consent for tracking and cookies. Any synced biometric data (heart rate, sleep data) from external APIs (e.g., Apple Health) must be encrypted at rest and in transit, following best-practice data privacy standards for health information.</i>	High
8	Other	<i>• Excited to try the new tools.</i> <i>• Highly invested and committed.</i> <i>• Feeling like a "VIP" user.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The application must launch quickly and stream workout media (videos/audio cues) without noticeable buffering to ensure a seamless exercise experience.</i>	<i>Initial app launch time under 2 seconds. Workout video buffering time under 1 second on standard 4G/LTE networks.</i>
2	Scalability	<i>The backend architecture (utilizing Supabase/MongoDB Atlas) must efficiently handle high spikes in concurrent active users, specifically during peak morning and evening workout hours.</i>	<i>System seamlessly supports 50,000+ concurrent active sessions with API response times remaining under 300ms.</i>
3	Security & Data Privacy	<i>All Personal Identifiable Information (PII) and synchronized biometric health data (heart rate, weight) must be strictly protected and anonymized for analytic</i>	<i>System seamlessly supports 50,000+ concurrent active sessions with API response times remaining under 300ms.</i>
4	Reliability & Availability	<i>Core tracking features, daily streak logs, and workout timers must remain functional consistently so users do not lose their progress.</i>	<i>99.9% guaranteed uptime for core services. Offline workout data must successfully sync 100% of the time upon internet reconnection.</i>
5	Usability & Accessibility	<i>The UI must be highly navigable during active physical movement, featuring large touch targets, high contrast modes, and text-to-speech support for visually impaired user</i>	<i>Achieves WCAG 2.1 AA compliance. 95% of beta testers can successfully initiate a workout within 3 taps from the home screen.</i>

6	Other	<i>The app must minimize smartphone battery consumption during lengthy workout sessions, especially when running background tasks like GPS tracking or Bluetooth wearable sync.</i>	App consumes less than 12% of total device battery per hour of active workout tracking on standard modern devices.
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